

DR DANTE KALISE

Associate Professor (Reader) in Computational Optimisation and Control
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RESEARCH AREAS: LEARNING, COMPUTATION, AND CONTROL

Foundations. Dynamic optimisation with ODE/PDE constraints; mathematical control theory; high-dimensional approximation and learning; numerical methods for Hamilton–Jacobi PDEs; optimal transport, mean-field control and games.

Applications. Scientific machine learning and data-driven control design; agent-based models, collective behaviour and consensus control; control of PDEs in fluid flow, vibration, and optimal actuator design.

APPOINTMENTS

Assoc. Prof. (Reader)	Dept. of Mathematics, Imperial College London	Since 9/2023
Mercator Professor	University of Wuppertal, Germany	Since 4/2025
Senior Lecturer	Imperial College London	7/2021–8/2023
ICL Research Fellow	Imperial College London	9/2017–5/2019
Assistant Professor	University of Nottingham, UK	6/2019–6/2021
Research Scientist	RICAM, Linz, Austria	5/2013–8/2017
Postdoctoral Researcher	Sapienza Univ. Rome / Univ. Bayreuth	11/2011–4/2013
PhD Researcher	Storm Weather Center, Norway	8/2008–10/2011

EDUCATION

PhD in Mathematics	University of Bergen, Norway	6/2012
MSc in Electronic Engineering	Federico Santa María University, Chile	7/2008
BSc in Engineering Mathematics	Federico Santa María University, Chile	12/2006

Also: Fellow of Advance HE (2020); Italian National Scientific Habilitation in Numerical Analysis (2018).

SELECTED FUNDED RESEARCH AWARDS

- **PI**, AFOSR–EOARD: *Optimal feedback recovery from high-dimensional, high-order data*, 2026–2030.
- **PI**, Imperial–Toyota Research Labs Visiting Researcher Agreement, 2025–2027.
- **CoI/Co-Director for Training**, EPSRC CDT in Collaborative Computational Modelling at the Interface (CCMI), since 2025.
- **CoPI**, EPSRC New Horizons: *Overcoming the Curse of Dimensionality with Tensor Decompositions*, 2021–2023.
- **PI**, Imperial College Research Fellowship: *Multiscale optimal control of collective behaviour*, 2017–2019.

SELECTED AWARDS

- 2022 Times Higher Education Award for Best STEM Research Project (with C. Wells, PhD student).
- Most-read paper in *Mathematical Models and Methods in Applied Sciences* (CBO via jump-diffusion SDEs).
- ITN-SADCO Postdoctoral Grant, European Union 7th Framework Programme (2011–2013).
- Bicentenary PhD Fellowship, Chilean Ministry of Education.

SELECTED PUBLICATIONS

Edited volumes: 3 (De Gruyter Radon Series 2018, 2022; Springer LNM 2017). Full list (~75 items, with preprints) at dkalise.net/publications.

8. Y. Huang, M. Herty, D. Kalise and N. Kantas. *Fast and robust consensus-based optimization via optimal feedback control*, SIAM J. Scientific Computing 48(1)(2026):A74–A102.
7. D. Kalise, A. Sharma and M.V. Tretyakov. *Consensus based optimization via jump-diffusion stochastic differential equations*, Math. Models Methods Appl. Sci. 33(2)(2023):289–339.
6. S. Dolgov, D. Kalise and K. Kunisch. *Tensor decomposition methods for high-dimensional Hamilton-Jacobi-Bellman equations*, SIAM J. Scientific Computing 43(3)(2021):A1625–A1650.
5. B. Azmi, D. Kalise and K. Kunisch. *Optimal feedback law recovery by gradient-augmented sparse polynomial regression*, J. Machine Learning Research 22(48)(2021):1–32.
4. D. Kalise and K. Kunisch. *Polynomial approximation of high-dimensional Hamilton-Jacobi-Bellman equations and applications to feedback control of semilinear parabolic PDEs*, SIAM J. Scientific Computing 40(2)(2018):A629–A652.
3. G. Albi, Y.-P. Choi, M. Fornasier and D. Kalise. *Mean field control hierarchy*, Applied Mathematics & Optimization 76(1)(2017):93–135.
2. G. Albi, M. Bongini, E. Cristiani and D. Kalise. *Invisible control of self-organizing agents leaving unknown environments*, SIAM J. Applied Mathematics 76(4)(2016):1683–1710.
1. A. Alla, M. Falcone and D. Kalise. *An efficient policy iteration algorithm for the solution of dynamic programming equations*, SIAM J. Scientific Computing 37(1)(2015):A181–A200.

EDITORIAL DUTIES

- Associate Editor: *Mathematics of Control, Signals, and Systems* (since 2019); *Networks & Heterogeneous Media* (since 2023); *Advances in Continuous and Discrete Models*, control section (since 2021).
- Associate Editor, SIAM book series *Advances in Design and Control* (since 2026).

LEADERSHIP, SERVICE, AND SUPERVISION

- Program Director, SIAM Activity Group on Control & Systems Theory (since 2026); News Liaison (since 2025).
- Course Director, MSc Applied Mathematics, Imperial College London (since 2023).
- **Supervision:** 8 PhD students supervised to completion and 9 in progress; 4 postdoctoral researchers; 35+ MSc/MMath projects.
- **Conference organisation:** co-organiser of SIAM CT27 and 15+ minisymposia and workshops; Programme Committee, MTNS (2022, 2024, 2026).
- Referee for EPSRC, DFG, AFOSR, ANR, ANID and 40+ journals; 17 PhD thesis committees.

INVITED TALKS (SELECTED, 2025–2026)

40+ invited talks since 2021, including plenary/keynote lectures. Recent: keynote at the COST Action InterCoML conference (Prague, 2026); plenary at the Brazilian Congress on Computational and Applied Mathematics (2025); invited talks at SIAM OP26 (Edinburgh), ECC 2026 (Reykjavík), MTNS 2026 (Waterloo), and the Oberwolfach workshop on Mathematical Foundations of Digital Twins (2026).